Transportation Programming Project - Phase II

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Abstract

This phase of the project develops and compares several trip distribution models for a multi–zone ridesharing network in Boston. Starting from the raw ride-level dataset, an origin–destination (OD) matrix was constructed for the entire study period and disaggregated by day of week to reveal temporal variations in spatial travel patterns. Flow balance analysis showed that a few zones (e.g., Back Bay, West End) act as net trip producers, while others (e.g., North End, Fenway, Boston University) are net attractors, implying non-negligible production–attraction imbalances that any forecasting model must respect. Using projected future trip productions and attractions for 12 zones, four families of distribution models were implemented in Python: the Furness iterative proportional fitting method, the Detroit and Fratar growth-factor methods, and gravity models with exponential and power impedance functions. Each algorithm was calibrated until either a 1% marginal-error tolerance or a maximum of 1000 iterations was reached, and the resulting OD matrices and convergence histories were analyzed. The results indicate that the gravity model with exponential impedance provides the best compromise between numerical stability, realism of flows, and computational efficiency, converging in under 70 iterations with low residual error and smooth, strictly positive OD entries. In contrast, Furness converges slowly, Detroit and Fratar exhibit poor growth-factor consistency, and the power-impedance gravity model produces overly polarized flows due to strong distance penalization. These findings justify adopting the exponential gravity formulation as the preferred basis for subsequent revenue-optimization and service-planning phases of the project.

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1 OD Matrix Analysis

1.1 Overall OD Matrix (All Days)

Using the raw rideshare dataset, we constructed a square origin–destination (OD) matrix T_{ij}, where each row i and column j correspond to one of the 12 traffic analysis zones, and entry T_{ij} represents the total number of trips from zone i to zone j over the whole observation period. The zones are: Financial District, Theatre District, Back Bay, Haymarket Square, Boston University, Fenway, North End, Northeastern University, South Station, West End, Beacon Hill, and North Station.

Table 8 reports the OD matrix aggregated over all days of the week.

Table 1: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	2395	2011	0	2138	2166	0	2210	2089	0	0
Beacon Hill	0	0	1928	2121	0	2144	2317	0	2103	1918	0	0
Boston Univ.	1870	2086	0	0	2245	0	0	2024	0	0	2082	2133
Fenway	2107	1965	0	0	2136	0	0	2078	0	0	2048	2017
Financial Dist.	0	0	2305	2257	0	2073	2043	0	2142	2162	0	0
Haymarket Sq.	2135	2125	0	0	2164	0	0	2016	0	0	2123	2036
North End	2081	2151	0	0	1922	0	0	2017	0	0	2102	1839
North Station	0	0	1912	2282	0	2076	2164	0	2125	1907	0	0
Northeastern Univ.	2040	2181	0	0	2056	0	0	2267	0	0	2186	1887
South Station	2064	2010	0	0	2342	0	0	1900	0	0	2109	2078
Theatre Dist.	0	0	2184	2024	0	2128	2236	0	2140	2144	0	0
West End	0	0	2325	2307	0	2057	1953	0	2224	1930	0	0

From Table 8, we see that all zones are strongly interconnected, and the largest flows typically occur between dense activity centers such as Back Bay, Boston University, Haymarket Square, and Northeastern University. Self-flows (diagonal elements) are zero by construction, since trips with identical origin and destination are not present in the dataset.

1.2 Day-of-Week OD Matrices

To investigate temporal patterns, separate OD matrices were constructed for each day of the week (Monday to Sunday). Each daily matrix aggregates trips only for that specific day while preserving the same 12-zone structure. This allows us to observe whether spatial travel patterns remain stable during the week or change between weekdays and weekends.

Tables **??**–**??** show the OD matrices for each day. For brevity, we report the full matrices but focus the discussion on key trends.

Table 2: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	462	358	0	349	400	0	368	386	0	0
Beacon Hill	0	0	297	382	0	421	430	0	365	312	0	0
Boston Univ.	326	369	0	0	390	0	0	385	0	0	339	377
Fenway	407	370	0	0	405	0	0	406	0	0	360	385
Financial Dist.	0	0	444	415	0	363	323	0	394	407	0	0
Haymarket Sq.	357	384	0	0	396	0	0	371	0	0	375	404
North End	356	381	0	0	326	0	0	357	0	0	363	351
North Station	0	0	384	392	0	362	359	0	369	364	0	0
Northeastern Univ.	403	389	0	0	317	0	0	423	0	0	381	361
South Station	335	371	0	0	426	0	0	336	0	0	387	372
Theatre Dist.	0	0	386	361	0	404	418	0	396	363	0	0
West End	0	0	416	443	0	371	355	0	402	338	0	0

Table 3: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	407	364	0	391	426	0	423	382	0	0
Beacon Hill	0	0	333	348	0	383	398	0	398	337	0	0
Boston Univ.	341	380	0	0	396	0	0	362	0	0	362	386
Fenway	383	317	0	0	362	0	0	360	0	0	430	389
Financial Dist.	0	0	434	425	0	352	407	0	420	392	0	0
Haymarket Sq.	394	340	0	0	371	0	0	405	0	0	389	341
North End	372	364	0	0	365	0	0	373	0	0	412	322
North Station	0	0	294	438	0	380	393	0	377	328	0	0
Northeastern Univ.	384	372	0	0	394	0	0	417	0	0	402	314
South Station	399	365	0	0	418	0	0	293	0	0	364	398
Theatre Dist.	0	0	410	375	0	407	385	0	362	434	0	0
West End	0	0	392	390	0	385	360	0	438	328	0	0

Table 4: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	270	189	0	274	267	0	234	202	0	0
Beacon Hill	0	0	209	233	0	218	267	0	214	201	0	0
Boston Univ.	206	249	0	0	248	0	0	206	0	0	231	246
Fenway	232	227	0	0	214	0	0	210	0	0	194	199
Financial Dist.	0	0	256	266	0	270	208	0	211	234	0	0
Haymarket Sq.	250	237	0	0	230	0	0	205	0	0	253	223
North End	244	240	0	0	196	0	0	216	0	0	248	186
North Station	0	0	224	234	0	213	240	0	204	222	0	0
Northeastern Univ.	237	222	0	0	240	0	0	228	0	0	264	195
South Station	206	249	0	0	253	0	0	229	0	0	200	207
Theatre Dist.	0	0	214	200	0	241	238	0	234	218	0	0
West End	0	0	269	260	0	245	210	0	223	204	0	0

Table 5: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	306	275	0	288	303	0	319	274	0	0
Beacon Hill	0	0	332	313	0	328	342	0	325	320	0	0
Boston Univ.	252	290	0	0	330	0	0	315	0	0	334	310
Fenway	315	290	0	0	335	0	0	309	0	0	271	276
Financial Dist.	0	0	310	309	0	289	322	0	315	325	0	0
Haymarket Sq.	293	319	0	0	329	0	0	280	0	0	286	304
North End	303	333	0	0	288	0	0	301	0	0	304	251
North Station	0	0	279	344	0	297	307	0	340	272	0	0
Northeastern Univ.	295	345	0	0	256	0	0	369	0	0	311	260
South Station	296	290	0	0	363	0	0	280	0	0	301	264
Theatre Dist.	0	0	319	337	0	265	302	0	289	338	0	0
West End	0	0	393	332	0	282	278	0	313	256	0	0

Table 6: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	355	250	0	267	265	0	297	271	0	0
Beacon Hill	0	0	252	281	0	283	272	0	263	274	0	0
Boston Univ.	264	266	0	0	284	0	0	275	0	0	282	253
Fenway	245	250	0	0	253	0	0	256	0	0	299	234
Financial Dist.	0	0	305	282	0	263	229	0	259	261	0	0
Haymarket Sq.	279	293	0	0	277	0	0	221	0	0	268	260
North End	259	249	0	0	241	0	0	240	0	0	255	260
North Station	0	0	230	300	0	258	277	0	291	272	0	0
Northeastern Univ.	254	285	0	0	283	0	0	280	0	0	296	210
South Station	253	239	0	0	291	0	0	253	0	0	301	268
Theatre Dist.	0	0	296	281	0	270	321	0	310	289	0	0
West End	0	0	268	261	0	272	257	0	268	282	0	0

Table 7: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	301	268	0	297	239	0	276	292	0	0
Beacon Hill	0	0	235	294	0	228	295	0	259	254	0	0
Boston Univ.	243	274	0	0	302	0	0	225	0	0	267	301
Fenway	261	266	0	0	252	0	0	279	0	0	226	242
Financial Dist.	0	0	290	294	0	267	290	0	233	260	0	0
Haymarket Sq.	277	283	0	0	300	0	0	284	0	0	287	254
North End	277	315	0	0	244	0	0	243	0	0	252	193
North Station	0	0	250	316	0	259	306	0	264	228	0	0
Northeastern Univ.	243	272	0	0	278	0	0	288	0	0	243	245
South Station	286	240	0	0	311	0	0	264	0	0	286	282
Theatre Dist.	0	0	290	221	0	250	278	0	251	235	0	0
West End	0	0	282	302	0	260	253	0	264	262	0	0

Table 8: OD matrix aggregated over all days

Source / Destination	Back Bay	Beacon Hill	Boston Univ.	Fenway	Financial Dist.	Haymarket Sq.	North End	North Station	Northeastern Univ.	South Station	Theatre Dist.	West End
Back Bay	0	0	294	307	0	272	266	0	293	282	0	0
Beacon Hill	0	0	270	270	0	283	313	0	279	220	0	0
Boston Univ.	238	258	0	0	295	0	0	256	0	0	267	260
Fenway	264	245	0	0	315	0	0	258	0	0	268	292
Financial Dist.	0	0	266	266	0	269	264	0	310	283	0	0
Haymarket Sq.	285	269	0	0	261	0	0	250	0	0	265	250
North End	270	269	0	0	262	0	0	287	0	0	268	276
North Station	0	0	251	258	0	307	282	0	280	221	0	0
Northeastern Univ.	224	296	0	0	288	0	0	262	0	0	289	302
South Station	289	256	0	0	280	0	0	245	0	0	270	287
Theatre Dist.	0	0	269	249	0	291	294	0	298	267	0	0
West End	0	0	305	319	0	242	240	0	316	260	0	0

Discussion. Comparing the daily OD matrices, we observe that the overall spatial pattern is relatively stable across weekdays: high flows consistently occur between dense central zones (e.g., Back Bay, Haymarket Square, Boston University, Northeastern University), while flows involving peripheral zones tend to be smaller. Weekend days (Saturday and Sunday) show slightly lower flows for many OD pairs, but the ranking of the busiest OD pairs remains similar, indicating that the structure of demand is largely preserved while the overall level of travel decreases.

1.3 Origin–Destination Balance

To assess whether each zone behaves as a purely transit node or as a net trip producer/attractor, we compared, for each zone, the total number of outgoing trips (sum over its row in the OD matrix) with the total number of incoming trips (sum over its column). Table 9 summarizes outgoing, incoming and the resulting balance

$$\text{balance}_i = \text{outgoing}_i - \text{incoming}_i.$$

Table 9: Outgoing, incoming and flow balance per zone (all days)

Zone	Outgoing	Incoming	Balance (Out – In)
Back Bay	13009	12297	712
Beacon Hill	12531	12518	13
Boston University	12440	13049	-609
Fenway	12351	13002	-651
Financial District	12982	12865	117
Haymarket Square	12599	12616	-17
North End	12112	12879	-767
North Station	12466	12302	164
Northeastern University	12617	12944	-327
South Station	12503	12150	353
Theatre District	12856	12650	206
West End	12796	11990	806

All zones exhibit some level of imbalance between outgoing and incoming flows. Back Bay and West End show the largest positive balances, indicating that they act as net trip *producers* (more departures than arrivals). In contrast, North End, Fenway and Boston University have negative balances, so they behave as net *attractors* of trips over the period. The magnitudes of the balances are moderate compared to the total flows, which means that, at the network level, the OD system is reasonably close to balanced, but local production/attraction effects are clearly present and should be considered in subsequent trip distribution modelling.

2 Trip Distribution Models

In this section, four alternative trip distribution models are calibrated on the base OD matrix and the projected future productions and attractions: (i) Furness (iterative proportional fitting), (ii) Detroit growth-factor method, (iii) Fratar growth-factor method, and (iv) gravity models with exponential and power impedance functions. All models produce a 12×12 OD matrix $T_{ij}^{(\cdot)}$, which is compared in terms of convergence behaviour and plausibility.

2.1 Furness Algorithm

The Furness algorithm iteratively scales rows and columns of an initial matrix $T_{ij}^{(0)}$ (here taken from the observed OD matrix) so that the row sums match the target origins O_i^{future} and the column sums match the target destinations D_j^{future}. At iteration k, row and column factors are given by

$$f_i^{(k)} = \frac{O_i^{\text{future}}}{\sum_j T_{ij}^{(k)}}, \qquad g_j^{(k)} = \frac{D_j^{\text{future}}}{\sum_i T_{ij}^{(k+1/2)}},$$

and the matrix is updated as

$$T_{ij}^{(k+1/2)} = f_i^{(k)} T_{ij}^{(k)}, \qquad T_{ij}^{(k+1)} = g_j^{(k)} T_{ij}^{(k+1/2)}.$$

Table 10 shows the resulting OD matrix after running the Furness procedure up to the prescribed maximum number of iterations.

Table 10: Predicted OD matrix using the Furness algorithm

Source / Destination	Financial Dist.	Theatre Dist.	Back Bay	Haymarket Sq.	Boston Univ.	Fenway	North End	Northeastern Univ.	South Station	West End	Beacon Hill	North Station
Financial Dist.	0.0000	0.0000	0.0000	9111.81	10264.19	9611.99	9088.82	9195.99	10197.39	0.0000	0.0000	0.0000
Theatre Dist.	0.0000	0.0000	0.0000	9600.73	9982.37	8847.48	10210.30	9430.19	10379.72	0.0000	0.0000	0.0000
Back Bay	0.0000	0.0000	0.0000	9532.22	10817.84	8687.10	9774.15	9623.94	9994.31	0.0000	0.0000	0.0000
Haymarket Sq.	9944.25	9625.43	10073.62	0.00	0.00	0.00	0.00	0.00	0.00	9516.31	9786.76	9128.99
Boston Univ.	10430.13	9543.54	8920.47	0.00	0.00	0.00	0.00	0.00	0.00	10079.53	9712.99	9266.19
Fenway	9998.24	9458.18	10126.50	0.00	0.00	0.00	0.00	0.00	0.00	9602.94	9218.29	9584.85
North End	9209.75	9937.62	10238.57	0.00	0.00	0.00	0.00	0.00	0.00	8962.98	10329.99	9523.97
Northeastern Univ.	9102.03	9548.18	9272.96	0.00	0.00	0.00	0.00	0.00	0.00	8496.95	9676.90	9889.73
South Station	11015.59	9787.06	9967.89	0.00	0.00	0.00	0.00	0.00	0.00	9941.28	9475.06	8806.27
West End	0.0000	0.0000	0.0000	9301.84	10651.39	10107.85	8938.63	9822.98	9365.26	0.0000	0.0000	0.0000
Beacon Hill	0.0000	0.0000	0.0000	9880.44	9001.34	9470.41	10807.16	9465.96	9484.80	0.0000	0.0000	0.0000
North Station	0.0000	0.0000	0.0000	9412.98	8782.87	10025.17	9930.95	9410.93	9278.52	0.0000	0.0000	0.0000

The Furness solution reallocates flows while preserving the general structure of the original matrix: zeros on many within-zone and cross-group cells remain zero, whereas large inter-zone flows are scaled up or down to satisfy the new marginal totals. However, the algorithm reached the maximum of 1000 iterations with a final relative error of about 1.1%, indicating slow convergence under the chosen tolerance.

2.2 Detroit Growth-Factor Method

The Detroit method uses base-year OD flows $T_{ij}^{(0)}$ and separate origin and destination growth factors to obtain a new matrix. At each iteration, growth factors for origins and destinations are updated as

$$g_i^{(k)} = \frac{O_i^{\text{future}}}{\sum_j T_{ij}^{(k)}}, \qquad h_j^{(k)} = \frac{D_j^{\text{future}}}{\sum_i T_{ij}^{(k)}},$$

and the flows are adjusted via

$$T_{ij}^{(k+1)} = g_i^{(k)}\, h_j^{(k)}\, T_{ij}^{(0)}.$$

Convergence is defined in terms of the growth factors falling within the range [0.99, 1.01].

The resulting OD matrix is reported in Table 11.

Table 11: Predicted OD matrix using the Detroit growth-factor method

Source / Destination	Financial Dist.	Theatre Dist.	Back Bay	Haymarket Sq.	Boston Univ.	Fenway	North End	Northeastern Univ.	South Station	West End	Beacon Hill	North Station
Financial Dist.	0.0000	0.0000	0.0000	2009.65	2263.81	2119.97	2004.58	2028.22	2249.08	0.0000	0.0000	0.0000
Theatre Dist.	0.0000	0.0000	0.0000	2117.48	2201.66	1951.35	2251.92	2079.87	2289.29	0.0000	0.0000	0.0000
Back Bay	0.0000	0.0000	0.0000	2102.37	2385.92	1915.98	2155.73	2122.60	2204.29	0.0000	0.0000	0.0000
Haymarket Sq.	2137.26	2068.74	2165.06	0.00	0.00	0.00	0.00	0.00	0.00	2045.28	2103.41	1962.04
Boston Univ.	2241.69	2051.14	1917.22	0.00	0.00	0.00	0.00	0.00	0.00	2166.33	2087.56	1991.53
Fenway	2148.86	2032.79	2176.43	0.00	0.00	0.00	0.00	0.00	0.00	2063.90	1981.23	2060.02
North End	1979.40	2135.83	2200.52	0.00	0.00	0.00	0.00	0.00	0.00	1926.36	2220.17	2046.93
Northeastern Univ.	1956.25	2052.13	1992.98	0.00	0.00	0.00	0.00	0.00	0.00	1826.20	2079.80	2125.54
South Station	2367.52	2103.48	2142.34	0.00	0.00	0.00	0.00	0.00	0.00	2136.62	2036.42	1892.68
West End	0.0000	0.0000	0.0000	2051.56	2349.21	2229.33	1971.45	2166.50	2065.55	0.0000	0.0000	0.0000
Beacon Hill	0.0000	0.0000	0.0000	2179.17	1985.28	2088.74	2383.56	2087.76	2091.91	0.0000	0.0000	0.0000
North Station	0.0000	0.0000	0.0000	2076.07	1937.10	2211.09	2190.31	2075.62	2046.42	0.0000	0.0000	0.0000

Although the Detroit method respects the base-year OD pattern more explicitly than Furness, the growth factors did not converge to the [0.99, 1.01] interval within 1000 iterations (final maximum deviation ≈ 3.65). This suggests that the combination of target margins and the original OD structure is relatively hard to reconcile through simple multiplicative growth factors.

2.3 Fratar Method

The Fratar method is another growth-factor approach that updates flows based on origin and destination factors computed from the previous iteration's row and column sums. In the implemented two-sided version, factors are applied to both dimensions:

$$g_i^{(k)} = \frac{O_i^{\text{future}}}{\sum_j T_{ij}^{(k)}}, \qquad h_j^{(k)} = \frac{D_j^{\text{future}}}{\sum_i T_{ij}^{(k)}}, \qquad T_{ij}^{(k+1)} = g_i^{(k)}\, h_j^{(k)}\, T_{ij}^{(k)}.$$

With the chosen initialization and inputs, the Fratar solution coincides numerically with the Detroit solution (Tables identical to Table 11). The algorithm reached the maximum of 1000 iterations with a relatively large final error of about 0.79, indicating poor convergence compared to the gravity model with exponential impedance.

Table 12: Predicted OD matrix using the Fratar method (numerically identical to Detroit)

Source / Destination	Financial Dist.	Theatre Dist.	Back Bay	Haymarket Sq.	Boston Univ.	Fenway	North End	Northeastern Univ.	South Station	West End	Beacon Hill	North Station
Financial Dist.	0.0000	0.0000	0.0000	2009.65	2263.81	2119.97	2004.58	2028.22	2249.08	0.0000	0.0000	0.0000
Theatre Dist.	0.0000	0.0000	0.0000	2117.48	2201.66	1951.35	2251.92	2079.87	2289.29	0.0000	0.0000	0.0000
Back Bay	0.0000	0.0000	0.0000	2102.37	2385.92	1915.98	2155.73	2122.60	2204.29	0.0000	0.0000	0.0000
Haymarket Sq.	2137.26	2068.74	2165.06	0.00	0.00	0.00	0.00	0.00	0.00	2045.28	2103.41	1962.04
Boston Univ.	2241.69	2051.14	1917.22	0.00	0.00	0.00	0.00	0.00	0.00	2166.33	2087.56	1991.53
Fenway	2148.86	2032.79	2176.43	0.00	0.00	0.00	0.00	0.00	0.00	2063.90	1981.23	2060.02
North End	1979.40	2135.83	2200.52	0.00	0.00	0.00	0.00	0.00	0.00	1926.36	2220.17	2046.93
Northeastern Univ.	1956.25	2052.13	1992.98	0.00	0.00	0.00	0.00	0.00	0.00	1826.20	2079.80	2125.54
South Station	2367.52	2103.48	2142.34	0.00	0.00	0.00	0.00	0.00	0.00	2136.62	2036.42	1892.68
West End	0.0000	0.0000	0.0000	2051.56	2349.21	2229.33	1971.45	2166.50	2065.55	0.0000	0.0000	0.0000
Beacon Hill	0.0000	0.0000	0.0000	2179.17	1985.28	2088.74	2383.56	2087.76	2091.91	0.0000	0.0000	0.0000
North Station	0.0000	0.0000	0.0000	2076.07	1937.10	2211.09	2190.31	2075.62	2046.42	0.0000	0.0000	0.0000

2.4 Gravity Models

2.4.1 Exponential Impedance Function

In the gravity model, flows are proportional to the product of origin and destination totals and a distance-decay (impedance) function:

$$T_{ij} = A_i B_j O_i^{\text{future}} D_j^{\text{future}} f(c_{ij}),$$

where c_{ij} is the average distance between zones i and j, $f(c_{ij})$ is an impedance function, and A_i, B_j are balancing factors calibrated iteratively to match the productions and attractions. For the exponential case we use

$$f(c_{ij}) = \exp(-\beta c_{ij}), \qquad \beta = 0.2,$$

with a balancing exponent $\alpha = 0.05$ in the implementation.

The resulting OD matrix for the exponential impedance model is given in Table 13.

Table 13: Gravity model OD matrix with exponential impedance

Source / Destination	Financial Dist.	Theatre Dist.	Back Bay	Haymarket Sq.	Boston Univ.	Fenway	North End	Northeastern Univ.	South Station	West End	Beacon Hill	North Station
Financial Dist.	6425.96	6159.28	6302.02	4538.66	426.48	644.56	4997.04	846.50	8148.92	6113.68	6672.30	6080.45
Theatre Dist.	6398.92	6133.37	6275.50	3196.33	1600.98	1883.31	3622.70	3129.75	7228.80	6087.96	6644.23	6054.87
Back Bay	6244.95	5985.79	6124.51	1876.85	5359.66	5219.99	1577.62	5952.52	1566.63	5941.48	6484.36	5909.18
Haymarket Sq.	5190.67	4206.66	2198.82	3974.70	5792.46	5475.61	4459.54	5273.04	4323.84	6167.15	4674.30	6677.10
Boston Univ.	699.41	2007.07	5099.25	6053.77	8822.37	8339.78	6792.22	8031.25	6585.55	1795.47	2438.79	1403.42
Fenway	788.78	2356.24	5230.79	5848.39	8523.05	8056.84	6561.79	7758.78	6362.13	2097.84	2881.38	1641.71
North End	5451.49	4008.56	1669.24	4572.80	6664.09	6299.56	5130.60	6066.51	4974.48	4829.64	3258.51	5504.96
Northeastern Univ.	814.34	3306.04	4902.16	5573.08	8121.84	7677.57	6252.89	7393.54	6062.64	1903.02	2606.00	1491.67
South Station	8173.41	5034.66	1661.98	4881.79	7114.40	6725.23	5477.28	6476.44	5310.62	3099.89	2280.05	2973.67
West End	6480.00	6211.09	6355.02	5997.97	2131.34	1923.71	5246.14	1649.67	2967.47	6165.10	6728.42	6131.59
Beacon Hill	6625.02	6350.09	6497.25	4076.77	3154.06	2847.84	3375.39	2983.03	2498.45	6303.08	6879.00	6268.82
North Station	6407.04	6141.16	6283.47	6248.88	1789.27	1656.00	5256.79	1388.96	2670.45	6095.69	6652.66	6062.56

This model converged in only 68 iterations with a final error of about 1.0%, significantly faster than Furness, Detroit, or Fratar. The resulting flows remain smooth and strictly positive for most OD pairs, thanks to the continuous exponential decay of impedance.

2.4.2 Power Impedance Function

As an alternative, a power impedance function was tested:

$$f(c_{ij}) = c_{ij}^{-\gamma}, \qquad \gamma = 1.5.$$

A higher γ increases the penalty for long-distance trips and concentrates flows on short-distance pairs. The same balancing procedure and parameters ($\alpha = 0.05$) were used.

Table 14 reports the resulting OD matrix.

Table 14: Gravity model OD matrix with power impedance ($\gamma = 1.5$)

Source / Destination	Financial Dist.	Theatre Dist.	Back Bay	Haymarket Sq.	Boston Univ.	Fenway	North End	Northeastern Univ.	South Station	West End	Beacon Hill	North Station
Financial Dist.	6.3e-14	1.15e-05	1.17e-05	1.75e-20	8.54e-57	3.66e-52	1.29e-08	1.12e-48	5.82e+04	1.06e-05	1.16e-05	1.43e-11
Theatre Dist.	7.96e-05	1.45e+04	1.47e+04	2.66e-23	2.60e-32	6.40e-29	2.29e-10	2.02e-18	5.11e+02	1.33e+04	1.46e+04	1.80e-02
Back Bay	7.96e-05	1.45e+04	1.47e+04	2.17e-35	1.61e-05	1.62e-04	4.91e-28	4.93e+02	6.06e-49	1.33e+04	1.46e+04	1.80e-02
Haymarket Sq.	7.33e-19	5.19e-18	2.05e-37	6.49e-22	2.93e-09	2.79e-09	2.85e-09	2.78e-09	1.38e-30	3.47e+03	5.45e-17	5.62e+04
Boston Univ.	1.65e-56	6.37e-34	5.85e-14	3.33e-09	1.50e+04	1.43e+04	1.46e+04	1.42e+04	7.07e-18	1.27e-35	5.59e-32	4.78e-45
Fenway	7.89e-55	1.48e-30	4.33e-12	3.33e-09	1.50e+04	1.43e+04	1.46e+04	1.42e+04	7.07e-18	1.63e-32	2.86e-28	2.96e-42
North End	2.20e-10	5.28e-12	1.09e-32	3.34e-09	1.50e+04	1.44e+04	1.47e+04	1.43e+04	7.10e-18	3.99e-05	3.94e-20	4.57e-05
Northeastern Univ.	6.11e-54	1.22e-22	1.23e-12	3.21e-09	1.45e+04	1.38e+04	1.41e+04	1.37e+04	6.83e-18	2.31e-33	2.73e-29	5.47e-43
South Station	5.97e+04	1.37e-11	3.88e-39	1.74e-14	7.86e-02	7.50e-02	7.66e-02	7.46e-02	3.71e-23	3.83e-26	1.41e-34	6.91e-33
West End	7.71e-05	1.41e+04	1.42e+04	1.28e+03	5.35e-28	8.60e-29	7.47e+02	1.18e-30	1.41e-36	1.29e+04	1.41e+04	1.74e-02
Beacon Hill	7.98e-05	1.46e+04	1.47e+04	1.18e-16	4.93e-21	5.31e-22	2.71e-13	3.02e-20	4.31e-41	1.34e+04	1.46e+04	1.80e-02
North Station	1.06e-06	1.94e+02	1.96e+02	5.56e+04	1.62e-32	7.28e-33	3.87e+01	6.22e-35	1.03e-40	1.78e+02	1.95e+02	2.40e-04

The power-impedance gravity model converged slowly (1000 iterations, final error $\approx 3.2\%$) and produces extremely polarized flows: many OD entries are effectively zero, while others concentrate very large volumes. This behaviour is consistent with the strong distance penalty implied by $\gamma = 1.5$ and suggests that, for this dataset and spatial scale, an exponential impedance provides a more realistic and numerically stable representation of trip decay.

2.5 Convergence Comparison

Table 15 summarizes the convergence performance of all four models.

Table 15: Convergence behaviour of trip distribution models

Model	Iterations	Final error / deviation
Furness	1000	final relative error ≈ 0.0109
Detroit	1000	max growth-factor deviation ≈ 3.65
Fratar	1000	final relative error ≈ 0.785
Gravity (exponential)	68	final relative error ≈ 0.0100
Gravity (power)	1000	final relative error ≈ 0.0317

Among the tested approaches, the gravity model with exponential impedance clearly exhibits the best trade-off between convergence speed and numerical stability. It converges in fewer than 70 iterations, achieves a low final error, and produces smooth, strictly positive OD flows. Furness also achieves a low error but requires the full 1000 iterations, while Detroit and Fratar fail to reach satisfactory convergence. The power-impedance gravity model strongly over-emphasizes short distances, leading to highly skewed flows and slower convergence, which makes it less suitable for the present application.

3 Price–Distance Optimization

In this section, we formulate and solve a route-selection problem that optimizes the trade-off between revenue and operating distance for a ridesharing service on the Boston multi-zone network. The analysis is based on the OD zoning system introduced in Parts I and II and uses aggregated price and distance information from both Lyft and Uber.

3.1 Data Preparation and Aggregation

Starting from the trip-level dataset, we first filtered trips to those occurring between the 12 study zones and grouped them by origin, destination, day of week, time of day, and cab type. For each OD–day–time combination, mean price and mean distance were computed separately for Lyft and Uber. These summaries were then merged to obtain a single comparison table, and the two providers were averaged to form unified metrics:

$$\text{avg\_price} = \frac{\text{avg\_price}_{\text{Lyft}} + \text{avg\_price}_{\text{Uber}}}{2}, \quad \text{avg\_distance} = \frac{\text{avg\_distance}_{\text{Lyft}} + \text{avg\_distance}_{\text{Uber}}}{2}.$$

The resulting dataset, stored as `comparison_df.csv`, contains one row per candidate route, with the following structure:

$$(\text{source}, \text{destination}, \text{dayofweek}, \text{time\_period}, \text{avg\_price}, \text{avg\_distance}).$$

Table 16 illustrates a small sample of this dataset over different days and night-time (0–6) for selected OD pairs.

Table 16: Sample rows from `comparison_df` (illustrative subset)

Source	Destination	Day of week	Time period	Avg. price	Avg. distance
Back Bay	Boston University	Friday	Night (0–6)	14.25	1.44
Back Bay	Boston University	Monday	Night (0–6)	14.01	1.44
Back Bay	Boston University	Thursday	Night (0–6)	14.02	1.44
Back Bay	Boston University	Tuesday	Night (0–6)	14.12	1.44
Back Bay	Boston University	Wednesday	Night (0–6)	13.98	1.44
Back Bay	Fenway	Friday	Night (0–6)	13.99	1.39

To focus on working days, weekend observations (Saturday and Sunday) were removed from `comparison_df`, leaving only Monday to Friday combinations.

3.2 Optimization Model Formulation

The goal of the optimization is to select a subset of candidate routes (defined by origin, destination, day of week, and time period) that maximizes profit while respecting a daily distance budget of 20 km. For each route r in `comparison_df`, we define:

- avg_price_r: average fare across providers,
- avg_distance_r: average trip distance,
- $\text{day}(r)$: the day of week (Monday–Friday),
- $\text{profit}_r = \text{avg\_price}_r - c \cdot \text{avg\_distance}_r$, with cost per distance unit $c = 1$.

A binary decision variable $x_r \in \{0, 1\}$ is introduced for each route, where $x_r = 1$ if the route is operated and $x_r = 0$ otherwise.

The optimization problem can be written as:

$$\max \sum_r \text{profit}_r \, x_r$$

subject to a distance budget for each working day d:

$$\sum_{r:\, \text{day}(r)=d} \text{avg\_distance}_r \, x_r \leq 20 \quad \forall d \in \{\text{Monday}, \ldots, \text{Friday}\},$$

and

$$x_r \in \{0, 1\} \quad \forall r.$$

This model was implemented in `Python` using the `pulp` library and solved with the CBC mixed-integer programming solver.

3.3 Solution and Selected Routes

The solver reported an optimal solution, with a total profit of approximately \$1714.61 and 100 selected routes across the five working days. Routes with high fares and moderate distances were favored, while long routes with relatively low fares were not selected.

Table 17 shows a representative subset of the selected OD–day combinations, together with their average price, distance, and resulting profit:

$$\text{profit} = \text{avg\_price}.$$

Table 17: Sample of selected routes in the optimal solution

Source	Destination	Day of week	Time period	Avg. price	Avg. distance
Back Bay	Haymarket Square	Monday	Night (0–6)	18.12	2.37
Back Bay	Haymarket Square	Thursday	Night (0–6)	18.28	2.40
Back Bay	Haymarket Square	Wednesday	Night (0–6)	18.20	2.36
Back Bay	North End	Friday	Night (0–6)	19.03	2.89
Back Bay	North End	Monday	Night (0–6)	19.28	2.83
Back Bay	North End	Thursday	Night (0–6)	19.36	2.88
Back Bay	North End	Tuesday	Night (0–6)	19.75	2.86
Back Bay	North End	Wednesday	Night (0–6)	19.63	2.88
Boston University	Financial District	Friday	Night (0–6)	24.37	4.54
Boston University	Financial District	Monday	Night (0–6)	24.19	4.54
Boston University	Financial District	Thursday	Night (0–6)	24.10	4.55
Boston University	Financial District	Tuesday	Night (0–6)	24.18	4.52
Boston University	Financial District	Wednesday	Night (0–6)	24.12	4.55
Boston University	North Station	Friday	Night (0–6)	19.45	3.43
Boston University	North Station	Monday	Night (0–6)	20.22	3.44
Boston University	North Station	Thursday	Night (0–6)	20.08	3.43
Boston University	North Station	Tuesday	Night (0–6)	20.54	3.43
		. . . (remaining selected routes omitted for brevity) . . .			

The solution selects routes that systematically connect high-demand pairs such as Back Bay–North End and Boston University–Financial District or North Station, especially in the night period, where fares are relatively high while distances remain moderate. This pattern is consistent with the intuition that profitable operations focus on dense urban corridors; at the same time, the 20 km daily constraint ensures that each day's operation remains within a realistic fleet capacity.

3.4 Daily Utilization and Distance Budget

To interpret how the 20 km budget is used per working day, we computed aggregate statistics of the selected routes by day of week. For each day d, the number of selected routes, the sum of their distances, and the utilization of the 20 km budget were calculated as

$$\text{Utilization}_d = \frac{\sum_{r:\,\text{day}(r)=d} \text{avg\_distance}_r \, x_r}{20} \times 100\%.$$

Table 18 reports the per-day utilization in the final solution. In all cases, the distance constraint is binding or nearly binding, ensuring efficient use of the daily capacity.

Table 18: Daily distance utilization under the optimal solution

Day of week	Selected routes	Total distance [km]	Utilization of 20 km
Monday	20	20.00	100%
Tuesday	20	20.00	100%
Wednesday	20	20.00	100%
Thursday	20	20.00	100%
Friday	20	20.00	100%

The results show that the model distributes the 100 selected routes evenly over the five working days, with 20 routes per day and full utilization of the 20 km distance budget each day. This symmetric allocation reflects the fact that both demand structure and price–distance trade-offs are relatively similar across weekdays in the aggregated dataset.

3.5 Discussion

The optimization in Part III demonstrates how aggregated OD-level price and distance information can be combined with a simple linear profit model and a daily distance constraint to design a compact but profitable service plan. The selected routes predominantly serve dense central OD pairs with high average fares and moderate distances, while routes with low profit or high distance are excluded. When integrated with the OD forecasts from Part II (especially the gravity model with exponential impedance), this framework provides a practical basis for revenue-oriented service planning, where the operator can adjust the daily distance budget, cost parameter c, or admissible time periods to explore different operational strategies.